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<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Jacobian Tutor</title>
  <script src="https://cdnjs.cloudflare.com/ajax/libs/mathjs/12.4.0/math.all.js"></script>
  <style>
    body { font-family: sans-serif; background: #f4f7f6; display: flex; justify-content: center;
padding: 20px; }
    .container { background: white; padding: 2rem; border-radius: 12px; box-shadow: 0 4px
15px rgba(0,0,0,0.1); max-width: 600px; width: 100%; }
    h1 { color: #2c3e50; text-align: center; }
    .problem-box { background: #eef2f3; padding: 15px; border-left: 5px solid #3498db;
margin: 20px 0; font-size: 1.2rem; }
    .step { margin-bottom: 20px; padding: 10px; border: 1px solid #ddd; border-radius: 8px; }
    input { padding: 8px; width: 80%; margin-top: 5px; border: 1px solid #ccc; border-radius:
4px; }
    button { background: #3498db; color: white; border: none; padding: 10px 20px;
border-radius: 5px; cursor: pointer; transition: 0.3s; }
    button:hover { background: #2980b9; }
    .feedback { margin-top: 10px; font-weight: bold; }
    .correct { color: #27ae60; }
    .incorrect { color: #e74c3c; }
  </style>
</head>
<body>

<div class="container">
  <h1>Jacobian Tutor</h1>
  <p>Find the Jacobian Matrix  $J$  for the following functions:</p>

  <div class="problem-box" id="problem">
     $f_1(x, y) = x^2 + y$  <br>
     $f_2(x, y) = 3x - y^2$ 
  </div>

  <div id="steps-container">
    <div class="step">
      <strong>Step 1:</strong> Find  $\frac{\partial f_1}{\partial x}$ 
      <br>
      <input type="text" id="ans1" placeholder="e.g., 2x">
      <button onclick="checkStep(1, '2x')">Check</button>
      <div id="feedback1" class="feedback"></div>
    </div>
  </div>
</body>

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</div>

<div class="step">
  <strong>Step 2:</strong> Find  $\frac{\partial f_1}{\partial y}$ 
  <br>
  <input type="text" id="ans2" placeholder="e.g., 1">
  <button onclick="checkStep(2, '1')">Check</button>
  <div id="feedback2" class="feedback"></div>
</div>
</div>

<button onclick="generateNew()">Generate New Problem</button>
</div>

<script>
  // Simple logic for checking answers using Math.js to compare expressions
  function checkStep(stepNum, correctAnswer) {
    const userInput = document.getElementById('ans' + stepNum).value;
    const feedback = document.getElementById('feedback' + stepNum);

    try {
      // Simplify both to see if they are mathematically equivalent
      const userSimp = math.simplify(userInput).toString();
      const correctSimp = math.simplify(correctAnswer).toString();

      if (userSimp === correctSimp) {
        feedback.innerHTML = "Correct! Well done.";
        feedback.className = "feedback correct";
      } else {
        feedback.innerHTML = "Not quite. Try again!";
        feedback.className = "feedback incorrect";
      }
    } catch (e) {
      feedback.innerHTML = "Invalid math expression.";
      feedback.className = "feedback incorrect";
    }
  }

  function generateNew() {
    alert("This is a template! You can expand the JavaScript to randomize f1 and f2.");
  }
</script>

</body>

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</html>